

SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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ABSTRACT

Fingerprinting Large Language Models (LLMs) is increasingly used for provenance verification and model attribution, but existing empirical evaluations concentrate on fine-tuned models where training data, optimization, and hyperparameters have already produced strong post-hoc signatures. Because most model capacity and knowledge are acquired during pretraining rather than downstream adaptation, the more demanding quantitative test for lineage verification is whether a fingerprint remains detectable from initialization through large-scale pretraining. We find that existing fingerprinting methods are *unreliable* under this test: their scores can fall below detection thresholds early in pretraining or under distribution shift, revealing that the measured signals emerge only after substantial training rather than at model birth.

We introduce **SeedPrints**, an intrinsic fingerprinting method that uses random-initialization bias as a persistent, seed-dependent identifier available before any training data are seen. Across random-input probes, untrained models exhibit reproducible prediction-bias profiles determined by the initialization seed; although these effects are weak in magnitude, they remain statistically detectable across training and support high-confidence lineage tests. Unlike prior techniques whose detection scores degrade in early checkpoints or after deployment-time changes, **SeedPrints** preserves the same comparison structure across initialization, large-scale pretraining, and downstream adaptation: same-seed lineages produce significant agreement, whereas different-seed baselines do not. Experiments on LLaMA-style and Qwen-style models show seed-level distinguishability, birth-to-lifecycle identity verification, and robust margins across prolonged training, domain shifts, and parameter modifications. Together, these results show that initialization itself imprints a unique and persistent identity on LLMs, providing a quantitatively verifiable “Galtonian” fingerprint. Code will be released upon acceptance.

1 INTRODUCTION

LLM fingerprints are intended to provide quantitative evidence for identifying, attributing, and tracing models from observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). In the standard comparison, a candidate or suspicious model is evaluated against a putative source model, and the method should return a stronger lineage score for true relatives than for unrelated baselines. Reliable separation in this setting gives model owners an empirical basis for detecting model theft or unauthorized reuse (Yoon et al., 2025; Zhang, 2025).

Much of this literature explicitly borrows the metaphor of biological fingerprints from Francis Galton’s *Finger Prints* (1892) (Galton, 1892):

“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”

Taken literally, this analogy imposes two empirical requirements: the signal should distinguish one identity from another, and the measured evidence should remain stable from a model’s “birth” at

SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

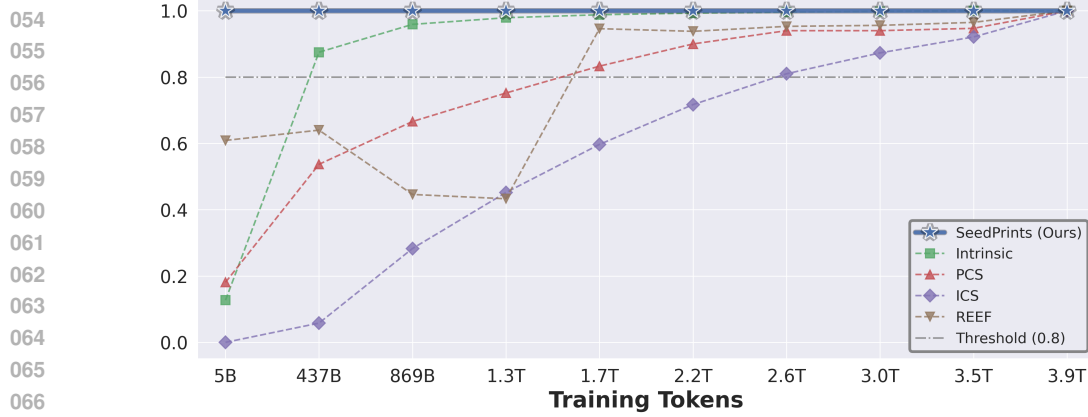


Figure 1: **Early-pretraining checkpoints expose a large detection gap between SeedPrints and existing fingerprinting baselines.** We compare five methods on OLMo-2-7B checkpoints from 5B to 3.9T training tokens, always testing each checkpoint against the final checkpoint. The y-axis reports the method-specific similarity score, where larger values indicate stronger lineage evidence; the dashed line marks the 0.8 detection threshold. All baselines lose margin and fall below threshold at early checkpoints, whereas **SeedPrints** remains at perfect detection ($p \ll 0.001$, plotted as $1 - p$) beginning with the first checkpoint.

initialization across its lifetime. Existing approaches often satisfy these criteria only after training has produced large, easy-to-measure artifacts (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). In Figure 1, this limitation appears quantitatively: early-pretraining checkpoints yield below-threshold evidence for all baselines, while the final-checkpoint comparison is easier. The same pattern reappears under training-distribution shift in Section 5.1, where baselines can misidentify lineage when data conditions change. These failures indicate that many current “fingerprints” are post hoc training identifiers rather than **Galtonian fingerprints**—intrinsic marks whose comparison margins remain meaningful from initialization onward.

We therefore evaluate fingerprinting under a stricter evidence standard: an LLM fingerprint should be present at *initialization* and remain statistically testable at *any* later training stage. Under this standard, our contributions are:

- **Quantifying a limitation of existing methods.** We show that prior fingerprinting scores are not consistently separated during early pretraining and can be confounded by substantial shifts in the training data distribution. (Section 5.1)
- **Turning initialization bias into a measurable fingerprint.** We find that untrained models already contain seed-dependent bias profiles in their internal representations, producing weak but reproducible identity evidence before training begins. (Section 3) We then introduce **SeedPrints**, which isolates these initialization-born signals and tests their agreement while reducing confounding from data distribution, training duration, and optimization dynamics. (Section 4)
- **Validating seed-level comparison margins.** We demonstrate that SeedPrints separates models differing only in initialization seed, even when their training pipeline and data order are identical, and that the same-seed signal persists after continual training on diverse datasets. (Section 5.1)
- **Showing practical significance across realistic evaluations.** SeedPrints detects lineage throughout large-scale pretraining trajectories such as OLMo-2-7B Stage 1 without the early-stage failures observed for baselines. It also remains competitive in large-scale fine-tuning settings, including up to 700B tokens, and on the *LeafBench* benchmark (Shao et al., 2025), which contains 65 models from 7 mainstream families and 6 transformations, including instruction tuning, fine-tuning, PEFT, quantization, model merging, and distillation. In these comparisons, SeedPrints matches the strongest near-perfect methods while outperforming weaker baselines by clear margins (Sections 5.2 and 5.3).

2 RELATED WORK

LLM protection methods differ in where their quantitative evidence comes from. One family, *watermarking / active fingerprinting*, inserts a measurable signature into a model or its outputs (Xu

et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024). The other, *passive fingerprinting*, extracts evidence from pre-existing model behavior without modifying the model (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025). SeedPrints belongs to the passive family, but differs in that its comparison signal is tied to initialization rather than to late-stage training artifacts.

Watermarking and Active Fingerprinting. Active approaches create ownership evidence by deliberately implanting verifiable identifiers before deployment. Classic mechanisms include backdoor triggers (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018), digital signatures, and hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, the main quantitative comparisons use either *text watermarks*, which bias generation or insert predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.), or *model weight watermarks*, which embed identifiers into parameters or tie them to secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Backdoor-style fingerprints can be straightforward to detect and may survive moderate fine-tuning (e.g., Dasgupta et al. 2024), but their empirical strength depends on invasive control of the training process. They therefore cannot retroactively provide evidence for third-party models that were not marked during training.

Passive LLM Fingerprinting. Passive fingerprinting instead compares models using signatures that already exist in weights, representations, or input-output behavior. With **white-box access**, methods score model pairs using properties such as attention-matrix distributions (Yoon et al., 2025), kernel alignment of internal representations (Zhang et al., 2024), or stable directions in parameter vectors. With **black-box access**, methods rely on behavioral evidence from crafted queries (Pasquini et al., 2024), distinctive prompt-response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025), or iterative prompting-response games (Iourovitski et al., 2024), although these scores can be less robust after fine-tuning. The shared limitation is that these signatures are usually defined *post hoc*: they measure properties that become prominent after training or alignment rather than properties already present at random initialization. Consequently, they are not optimized for the harder comparison regime studied here—lineage verification during large-scale pretraining, before post-training signals have fully emerged. SeedPrints targets this gap by testing an initialization-born signal and reporting calibrated statistical evidence for same-lineage agreement.

3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

This section quantifies the core empirical pattern behind SeedPrints: random initialization creates seed-specific prediction-bias profiles, and the same profiles remain measurably separated from independent-seed baselines after training.

Initialization induces seed-specific bias profiles. We first evaluate a LLaMA-2-style model initialized with seed 123 on 10,000 uniformly sampled random input sequences of length 1,024. In Figure 2 (Left), the plotted statistic is the minimum-attainment frequency for each output token (upper, final logits) and each hidden dimension (lower, last-layer representations) across the same 10,000 trials. The large deviation from the uniform reference line makes the bias visible: a small subset of dimensions is repeatedly disfavored far more often than a uniform output model would predict. Repeating the same measurement across initialization seeds preserves the overall bias magnitude but changes which dimensions occupy the argmin ranks (Figure 2, Lower Right), giving the comparison a seed-specific structure rather than a generic architectural artifact. These observations are consistent with recent findings that biases in randomly initialized models arise from inter-sequence representation contraction, driven by asymmetric nonlinear activations in MLP blocks and further amplified by self-attention (Li et al., 2026). Under this view, each seed selects different contraction directions, which explains why the empirical argmin pattern differs across output dimensions even when architecture and probing distribution are fixed.

Training preserves initialization-born bias Training changes output magnitudes substantially, but the ranking evidence attached to initialization-born output preferences does not disappear. To measure persistence, we train the seed-123 model for one epoch on OpenWebText (Gokaslan et al., 2019) and evaluate intermediate checkpoints against the initialized reference. The comparison is restricted to the $m = 50$ output dimensions most frequently assigned minimum values by the initialized model. For each checkpoint, we compute its random-input correlation with the initialized model on these dimensions, directly measuring whether the trained checkpoint preserves the same rank-level disfavoring pattern. The negative baseline uses the identical correlation statistic against

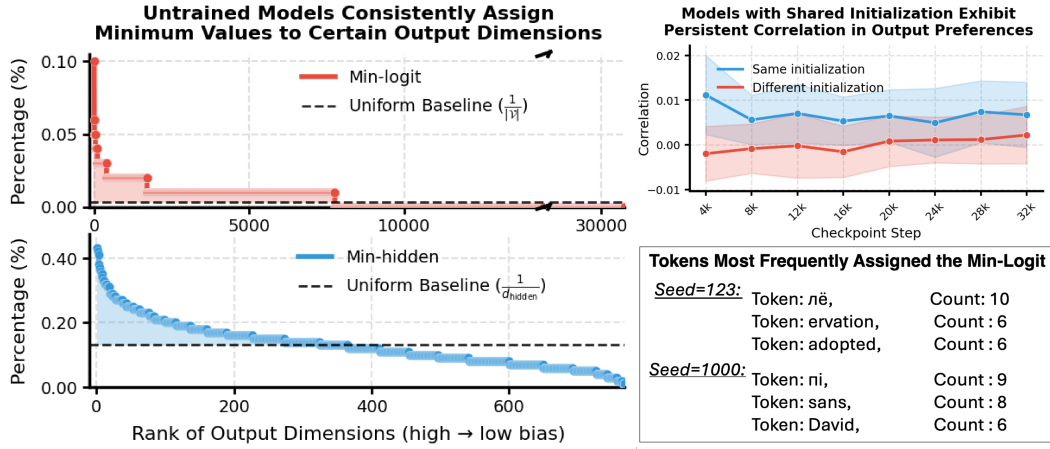


Figure 2: **Initialization-born output bias remains quantitatively detectable after training.** *Left:* Under completely random inputs, a randomly initialized LLaMA-2-style model shows a strong departure from the uniform-frequency baseline: selected dimensions receive the minimum value far more often than expected. The same extreme-bias pattern is visible in logits (top, red) and final hidden representations (bottom, blue). The dashed line marks the uniform expectation, and the arrows in the top panel indicate a broken x-axis omitting low-frequency tail ranks. *Upper Right:* Across training checkpoints, same-lineage correlations with the initialization reference remain consistently higher than correlations with independently initialized baselines, preserving a measurable margin despite small absolute values. *Lower Right:* The high-bias dimensions are seed-specific; the displayed tokens are those that most frequently receive the minimum logit for each initialization seed.

an independently initialized seed-1000 model. As shown in Figure 2 (Upper Right), the absolute correlations are small, but the same-initialization distribution is consistently shifted upward relative to the independent-seed baseline and stabilizes instead of decaying to zero. This persistent margin supplies the quantitative evidence that initialization leaves a weak but statistically detectable signal throughout training.

4 ALGORITHM

Section 3 shows that same-lineage models retain a small but consistent advantage over independent baselines on a subset of output dimensions, especially those disfavored by the untrained model under random inputs. SeedPrints turns this empirical margin into a testable comparison: for any two models, without access to the true initialized model, it extracts candidate bias dimensions and asks whether their agreement is significantly larger than the null expectation. The overall procedure is summarized in Algorithm 1, and we describe each component below.

Extract identity dimensions between any two models. Let $X = \{x_i\}_{i=1}^n$, where each $x_i \in \mathbb{R}^{\ell \times d}$ denotes a random sequence of length ℓ . The probing distribution is deliberately controlled: each x_i is instantiated either by uniformly sampling ℓ token IDs from the vocabulary or by directly sampling ℓ independent vectors from a d -dimensional isotropic Gaussian distribution. For any model g , we define its mean response vector as $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the output dimensionality. Here, $g(x_i)$ can denote either the model’s logits or its final-layer representations. We then convert this average-response estimate into an ordered fingerprint candidate by selecting the bottom- m coordinates of $\bar{g}$:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

The mean response vector is used because it yields a more stable empirical estimate than per-sample argmin frequency, i.e., counting how often each dimension attains the minimum value as in Figure 2. This choice improves noise robustness while preserving strong empirical performance; Appendix C.1 gives the formal comparison.

Given two models f and f' , we define their *identity dimensions* as the intersection $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$. For a same-initialization pair, both models should select overlapping low-mean dimensions more often than unrelated pairs, creating a first-stage overlap margin before any correlation statistic is computed. The intersection therefore acts as mutual verification: it reduces spurious agreement from

Algorithm 1 Distribution Correlation Test on Identity Dimensions

Require: base model f , suspicious model f' ; random inputs $X = \{x_i\}_{i=1}^n$; fingerprint size m ; significance level α

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs $\bar{f}, \bar{f}'$ over X
- 2: $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$ bottom- m dimensions of $\bar{f}, \bar{f}'$
- 3: $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$ (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$ for each $x_i \in X$ across identity dimension
- 5: **for** each $s_j \in \mathcal{S}$ **do**
- 6: $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$ (correlation in in input-wise preferences)
- 7: **end for**
- 8: $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{T}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct $\mathcal{T}_{\text{null}}$ by applying the same pipeline to two Gaussian matrices $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$ vs. $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

shared architecture or training dynamics and retains dimensions whose evidence is more likely to reflect shared initialization. When one model is closer to initialization, its disfavored dimensions can also serve directly as the comparison anchor.

Correlation statistics. After restricting both models to the identity dimensions $\mathcal{S}$, we compare their responses across the same random inputs.¹ For each $j \in \mathcal{S}$, we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

The resulting statistic set $T = \{\tau_j\}_{j \in \mathcal{S}}$ summarizes the full distribution of agreement across identity dimensions rather than a single heuristic similarity score. As observed in Section 3, same-lineage pairs shift this distribution upward relative to unrelated pairs, and SeedPrints evaluates that shift directly.

Hypothesis testing via null distribution. Because same-lineage and different-lineage correlation distributions can overlap, an individual-correlation threshold would discard useful aggregate evidence. SeedPrints instead tests whether the observed distribution is significantly shifted above the null hypothesis of no lineage relation. The ideal null would be estimated from many independent-model comparisons, as in Figure 2, but this empirical baseline is computationally expensive. Appendix B shows that the independent-initialization null is well approximated by a Gaussian distribution, motivating a cheaper simulation-based alternative that constructs an empirical null from Gaussian surrogates passed through the same pipeline, as summarized in Algorithm 1.

Although the simulation-based null is already a strong empirical approximation, it adds avoidable variance and overhead: precision depends on the number of trials, results may vary across runs, and empirical-distribution comparisons require assumptions from tests such as the one-sided t-test or the Mann–Whitney U test. Because the Kendall–Tau null distribution is available in closed form under weak independence, we replace the simulation step with an analytical test while also providing both empirical and analytical implementations in our code repository. Specifically, under the null hypothesis, each Kendall–Tau correlation is approximately zero-mean with known variance σ^2 . By the central limit theorem, as the number of identity dimensions $|\mathcal{S}|$ increases, the distribution of $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$ is well approximated by $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$. Softmax normalization creates weak dependencies across dimensions, but the relatively high temperature ($T = 10$) reduces these dependencies enough for the approximation to match the full-pipeline null empirically (Appendix B). Thus the final evidence score is a one-sided z-test: we compute $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$ and $p = 1 - \Phi(z)$, where Φ is the standard normal cumulative distribution function. We reject the null and declare shared lineage when $p < 0.01$.

¹Before computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This removes scale differences across different models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is used to avoid degenerate one-hot behavior.

Table 1: Cross-seed Seed-Prints comparisons: different initialization seeds produce non-significant lineage evidence (p -value > 0.01). Table 2: Same-seed initialization-to-trained comparisons: trained descendants retain significant SeedPrints evidence (p -value < 0.01).

Model Pairs		SeedPrints	Baselines			
		p -value	Intrinsic	REEF	PCS	ICS
s_{42} VS. s_{2000}	0.116	s_{42}^{init} vs. s_{42}^{base} 3.37e-26✓	-0.021×	0.375×	0.580×	0.196×
s_{123} VS. s_{42}	1.000	s_{123}^{init} vs. s_{123}^{base} 3.16e-33✓	0.149×	0.369×	0.581×	0.188×
s_{1000} VS. s_{123}	0.327	s_{1000}^{init} vs. s_{1000}^{base} 1.12e-21✓	-0.252×	0.381×	0.581×	0.188×
s_{2000} VS. s_{1000}	1.000	s_{2000}^{init} vs. s_{2000}^{base} 3.70e-31✓	-0.337×	0.331×	0.581×	0.188×

Practical instantiation. Model outputs can be taken as either logits or hidden representations, so the same test applies at different representation levels. In practice, logits require many more random inputs as vocabulary size grows, increasing the sample and compute cost needed to estimate bias patterns reliably (see Appendix C.2 for theoretical analysis). We therefore use final-layer hidden states by default, which gives more stable estimates at lower probing cost. The output is a calibrated p -value for lineage evidence rather than a heuristic similarity threshold.

5 EXPERIMENTS

The experiments are organized around two quantitative questions. First, in Section 5.1, we test whether SeedPrints provides *birth verification at the seed level* and whether the same evidence remains *verifiable across the entire training lifecycle*. These comparisons explicitly separate same-seed descendants from different-seed baselines and show where existing methods lose margin: early pretraining and large distribution shifts. Unless otherwise stated, we report SeedPrints with the analytical null and *hidden state* outputs; Appendix D.2 reports the corresponding empirical-null results for both *logit* and *hidden state* outputs using the one-sided t -test and Mann–Whitney U test. Second, in Section 5.3, we compare all methods in practical infringement scenarios using *LeafBench* (Shao et al., 2025). This benchmark contains 65 distinct model instances across 6 representative post-development techniques, allowing us to measure whether SeedPrints remains competitive with the strongest post-training baselines while preserving its initialization-level interpretation.

Baselines We compare against four passive weight- or representation-based fingerprinting baselines. *Intrinsic fingerprint* (Yoon et al., 2025) (or *PDF* in some papers) scores a pair of models by the similarity of layerwise standard-deviation profiles of attention parameters. *REEF* (Zhang et al., 2024) scores pairs using centered-kernel-alignment (CKA) similarity between feature representations on matched samples. *PCS* and *ICS* (Zeng et al., 2024) (collectively called *HuRef* in some papers) are weight-similarity baselines: PCS computes cosine similarity after flattening all parameters, whereas ICS computes cosine similarity on invariant weight terms. In Section 5.3, we also include the weaker gradient-based fingerprint *Gradient* (Shao et al., 2025). Following Zhang et al. (2024), baseline binary decisions use a **0.8 similarity threshold**.

In all experiment tables, cell color fixes the comparison condition: green marks same-source or same-lineage pairs, and red marks different-source pairs. For example, s_{42}^{init} vs. s_{42}^{base} compares the seed-42 initialization with its continued-pretrained descendant, whereas s_{2000}^{init} vs. s_{42}^{base} compares a seed-2000 initialization against a model trained from seed 42. The symbols ✓ and × denote correct and incorrect detection outcomes, respectively.

5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch so that the only controlled identity variable is the initialization seed. The main text reports the LLaMA-style results, and Appendix D reports the Qwen-style counterpart; the same empirical pattern holds in both families.

Different initialization seeds produce distinct fingerprints Table 1 reports SeedPrints p -values for independently initialized model pairs with seeds 42, 123, 1000, and 2000. Every cross-seed comparison remains above the p -value decision level of 0.01, so the method does not falsely merge

Table 3: Identical dataset and training order do not erase seed-level separation across different initializations.

Model Pair	p -value (> 0.01)
s_{42}^{init} vs. s_{123}^{base}	0.573
s_{123}^{init} vs. s_{2000}^{base}	0.724
s_{2000}^{init} vs. s_{1000}^{base}	0.109
s_{1000}^{init} vs. s_{42}^{base}	0.883

Table 4: Continual-training lineage tests under domain shift (base model: seed 1000, corpus openwebtext).

Setting	SeedPrints	Baselines			
Continual corpus (seed)	p -value	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	≈ 0 ✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	0.950×	0.658✓	0.332✓	0.012✓
the_stack (1000)	≈ 0 ✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	1.000✓	0.445✓	0.580✓	0.301✓	0.026✓

distinct seeds. This establishes the negative side of birth-level evidence: different random seeds yield separable fingerprint behavior before any training data are introduced.

Training preserves the initialization fingerprint. Table 2 tests the positive side of the same comparison by pairing each initialization model s^{init} with its OpenWebText-trained descendant s^{base} (Gokaslan et al., 2019) (≈ 10 B tokens). For every seed–descendant pair, SeedPrints returns $p < 0.01$, showing that the trained model remains statistically tied to its initialization. The baseline columns provide the contrast: across these seed-controlled pairs, they fail to recover the same birth-to-base relation, indicating that their separability is dominated by training-induced artifacts rather than initialization evidence.

Identical data and order do not make fingerprints converge In Table 3, all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus* (OpenWebText) and *in the same data order*; the training seed is fixed to lock the data order, leaving initialization seed i as the only intended difference. Across all cross-seed comparisons, SeedPrints returns $p > 0.01$, sharply contrasting with the near-zero same-seed values in Table 2. Thus identical data and curriculum do not collapse the seed-specific margins.

Continual training on diverse datasets does not confound the fingerprint The preceding seed-controlled experiments establish that SeedPrints behaves like a biometric identity signal, but seed separation alone is not the main copyright claim because initialization seeds themselves have unclear copyright status. The more practically important comparison is whether an attribution method still recognizes true lineage when the descendant is trained on a substantially shifted data distribution. This is where data-dependent baselines become fragile: their scores can track corpus similarity more strongly than lineage.

In Table 4, the source is a seed-1000 base model pretrained on OpenWebText (Gokaslan et al., 2019). We continue this base on two distributionally different corpora—TinyStories (Eldan & Li, 2023), consisting of synthetic children’s stories, and The Stack (Kocetkov et al., 2022), consisting of permissively licensed GitHub code—and compare true descendants against *distractors* derived from a seed-123 base. Because the baselines are not seed-sensitive, the distractor base is pretrained with a different OpenWebText data order and then continually trained on the *same* target corpus as the true descendant. The resulting test asks whether each method ranks actual weight lineage above target-corpus similarity.

Under the code-domain setting (The Stack), all prior baselines misclassify the true descendants as distractors, revealing that their evidence follows domain similarity rather than lineage identity. TinyStories is closer to OpenWebText, whereas The Stack is a much sharper distribution shift; the latter shift is sufficient to erase the baseline lineage margin. SeedPrints correctly attributes the source across both corpora, showing that its quantitative signal is not a proxy for data distribution and remains usable beyond the initial pretraining stage.

5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. We first ask whether SeedPrints identifies descendants at every point along a controlled training trajectory. In an OpenWebText pretraining run with known initialization seed, we test intermediate checkpoints against their lineage reference and observe consistent detection at all stages (Appendix D.2.1, Figure 5).

Large-scale pretraining. To evaluate lineage under realistic foundation-model training, we analyze ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run, spanning 5B $\rightarrow$ 3.9T

Table 5: Fingerprinting results under large-scale finetuning. Each row compares a target model against LLaMA-2-7B. **SeedPrints** reports the p -value from our correlation test (< 0.01 indicates a strong signal). Four baselines all report similarity scores (threshold = 0.8, higher = better).

Model	# Tokens	SeedPrints p	Intrinsic $\uparrow$	REEF ($\uparrow$)	PCS ($\uparrow$)	ICS ($\uparrow$)
Llama-2-finance-7B (Heenan, 2023)	5M	10^{-41955} ✓	1.0000✓	0.9950✓	0.9979✓	0.9952✓
Vicuna-1.5-7B (Chiang et al., 2023)	370M	$10^{-103043}$ ✓	1.0000✓	0.9985✓	0.9985✓	0.9949✓
Wizardmath-7B (Luo et al., 2023)	1.8B	$10^{-180550}$ ✓	1.0000✓	0.9979✓	1.0000✓	0.9994✓
Meditron-7B (Chen et al., 2023)	48B	10^{-42566} ✓	0.9990✓	0.9978✓	1.0000✓	0.9817✓
CodeLlama-7B (Meta AI, 2024)	500B	10^{-3552} ✓	0.9480✓	0.9947✓	0.6863✗	0.3369✗
Llemma-7B (Azerbayev et al., 2023)	700B	10^{-5136} ✓	0.9470✗	0.9984✓	0.6682✗	0.2905✗

tokens. The final checkpoint is treated as the target model and each earlier checkpoint as a tested ancestor. This setting is more demanding than controlled OpenWebText training because it includes heterogeneous corpora, longer optimization, and large representation drift. For visual comparability with similarity baselines, we plot $1 - p$; the underlying SeedPrints results have $p \ll 0.01$ and are numerically close to zero for all tested checkpoints.

In Figure 1, the baseline margin collapses during early pretraining: within the first **trillion** tokens, ICS is approximately 0, PCS is below 0.3, and REEF is below 0.6, all under the 0.8 detection threshold. SeedPrints, by contrast, remains above threshold from the first checkpoint and strengthens as training proceeds. This pattern identifies a critical evaluation gap: post-fine-tuning verification is substantially easier than early-pretraining verification, and late-stage success can give a *false sense of safety*. Lineage auditing should therefore emphasize early checkpoints, where misuse is hardest to detect and where the baseline failure margins are largest.

Large-scale finetuning. We also compare SeedPrints with existing baselines under standard fine-tuning evaluations. The suspect models are all fine-tuned from Llama-2-7b, with data volumes ranging from 5 million to 700 billion tokens: Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), CodeLlama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023). Their respective fine-tuning volumes are 5M, 370M, 1.8B, 13B, 500B, and 700B tokens. As shown in Table 5, SeedPrints keeps the lineage decision significant ($p < 0.01$) in every setting, including the largest-token fine-tuning cases.

5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world deployments routinely apply *parameter-altering* adaptations to foundation models, including fine-tuning, PEFT, and quantization, so a useful fingerprint must preserve separation after such transformations. We evaluate this practical robustness with *LeaFBench* (Shao et al., 2025), a copyright-auditing benchmark that compares derivative models against independent models under realistic post-development changes.

Across Source Models. We audit two source-model conditions: *Pre-Trained (PT)* sources and *Instruction-Tuned (IT)* sources. For each condition, the metric compares scores for *derivative* models, produced by post-training adaptations, against *independent* models trained without the source weights. This mirrors common lineage disputes involving either a PT base or its IT variant. The evaluation contains **65** models in total; Appendix E lists the model names and HuggingFace repositories.

Metrics. We report two complementary measures of distributional separation: (i) Area Under the ROC Curve (AUC), which summarizes ranking quality over thresholds, and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), which measures the maximum gap between derivative and independent score distributions.

SeedPrints outputs per-pair p -values, whereas baselines output similarity scores. This difference matters quantitatively: similarity scores are linear ranking quantities, but p -values are tail probabilities under the no-lineage null and are not linear in evidential strength. To compute AUC in the same format as the baselines, we convert p -values to scores using $s = 1 - p$. This conversion is conservative for SeedPrints because very small p -values vary on an exponential scale while AUC threshold sweeps are linear, compressing fine-grained evidence in the most significant region. Even under this imperfect conversion, SeedPrints remains comparable to or better than the strongest baselines. For evaluation, we compute AUC for all methods by sweeping thresholds over s and report KS as a threshold-free

Table 6: Source-model comparison on LeaFBench. “PT” and “IT” denote pre-trained and instruction-tuned source models, respectively; per-family entries report family-specific separability, while “Overall” is computed over all test model pairs rather than averaged across families.

Method	Metric	Qwen2.5-7B		Qwen2.5-14B		Llama3.1-8B		Mistral-7B		Gemma2-2B		Llama2-7B		Overall
		PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	
REEF	AUC $\uparrow$	0.796	0.862	0.947	0.913	1.000	0.999	0.997	0.997	0.963	0.968	0.750	0.706	0.915
	KS Statistic $\uparrow$	0.492	0.633	0.875	0.735	0.987	0.987	0.975	0.980	0.950	0.837	0.750	0.694	0.739
Gradient	AUC $\uparrow$	0.657	0.743	0.763	0.789	0.897	0.904	0.872	0.682	1.000	1.000	0.650	0.768	0.801
	KS Statistic $\uparrow$	0.354	0.423	0.500	0.563	0.646	0.705	0.725	0.500	1.000	1.000	0.275	0.479	0.508
ICS	AUC $\uparrow$	1.000	1.000	0.997	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	0.975	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.943
Intrinsic	AUC $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.989
SeedPrints	AUC $\uparrow$	0.992	0.994	0.988	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992
	KS Statistic $\uparrow$	0.985	0.987	0.975	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.986

distribution-separation measure. We omit Manhattan distance because absolute distances between p-values do not have a meaningful linear interpretation.

In most practical audits, an optimal similarity threshold cannot be calibrated in advance. AUC is therefore useful for benchmark comparability, but SeedPrints supplies a different operational output: **given any pair of models, it returns a p -value that supports a fixed-significance lineage decision without tuning a dataset-specific threshold.**

Parameter-Altering Techniques. To assess robustness by transformation type, we evaluate six parameter-altering techniques: (1) Instruction tuning (Instruct), (2) General-purpose fine-tuning (Finetune), (3) Parameter-efficient fine-tuning (PEFT), (4) Quantization, (5) Model merging (Merge), and (6) Distillation.

5.3.1 FAMILY-WISE EFFECTIVENESS

The benchmark covers seven model families. For each family, we treat the pretrained base model (PT) and the instruction-tuned model (IT) as separate sources and compare each source against all other models in the 65-model pool. Table 6 reports these results, omitting the TinyLlama-1.1B families because they are trivial for all methods. Each family column reports AUC under tests for that family using a family-calibrated threshold, whereas the overall score is computed across all test pairs with a global threshold; therefore, the overall score is not the average of per-family scores. The strongest group consists of *Intrinsic*, *ICS*, and *SeedPrints*, with overall AUCs of 0.994, 0.994, and 0.992 and KS values of 0.989, 0.943, and 0.986, respectively. Most family-wise entries for these methods are saturated, while *REEF* and *Gradient* lag by substantially larger margins. Thus, although SeedPrints is designed around intrinsic seed-level evidence, it also functions as a practical biometric-like fingerprint across diverse model families.

5.3.2 ALTERATION-WISE ROBUSTNESS

By alteration type, *Instruct* yields the clearest separation: even Gradient, the weakest baseline, reaches AUC 0.895. *Finetune* and *Merge* are harder for all methods; for example, no method is perfect on *Finetune*. The middle cases show method-specific margins: REEF drops under *Quantization* ($0.996 \rightarrow 0.871$) and *Distillation* ($0.996 \rightarrow 0.858$) while remaining strong on *PEFT* (0.994), whereas Gradient is weaker on *PEFT* and *Quantization* ($0.895 \rightarrow 0.776/0.756$). The three strongest methods remain above 0.9 across transformations. SeedPrints records its main drop on *Merge*, reaching 0.959 and missing one merge model out of seventeen; this isolated error suggests that interpolation with unrelated weights can dilute initialization evidence below the decision threshold. Averaged across all six alterations, Intrinsic has the highest overall AUC, with ICS and SeedPrints close behind.

5.4 ABLATION STUDY ON HYPERPARAMETERS

SeedPrints has three hyperparameters that control evidence quality and sample cost: the number of random input sequences n , the length of each random sequence l , and the fingerprint size m . The first two determine how much random-input evidence is collected, while m determines how many low-mean coordinates enter the lineage test. In the main experiments in Sections 5.2 and 5.3, we use

Table 7: Alteration-wise robustness of LLM fingerprinting methods, comparing derivative-versus-independent separability after each parameter-altering technique.

Method	Metric	Instruct	Finetune	PEFT	Quantization	Merge	Distillation	Overall Avg
REEF	AUC $\uparrow$	0.996	0.872	0.994	0.871	0.954	0.858	0.924
	KS Statistic $\uparrow$	0.949	0.637	0.940	0.812	0.851	0.615	0.801
Gradient	AUC $\uparrow$	0.895	0.729	0.776	0.756	0.832	0.887	0.812
	KS Statistic $\uparrow$	0.756	0.441	0.469	0.586	0.519	0.679	0.575
ICS	AUC $\uparrow$	1.000	0.987	1.000	1.000	0.981	1.000	0.995
	KS Statistic $\uparrow$	1.000	0.925	1.000	1.000	0.941	1.000	0.978
Intrinsic	AUC $\uparrow$	1.000	0.980	1.000	1.000	1.000	1.000	0.997
	KS Statistic $\uparrow$	1.000	0.971	1.000	1.000	1.000	1.000	0.995
SeedPrints	AUC $\uparrow$	1.000	0.995	0.990	1.000	0.959	1.000	0.991
	KS Statistic $\uparrow$	1.000	0.990	0.980	1.000	0.941	1.000	0.985

$n=2000$, $l=1024$, and $m=0.1d_{\text{out}}$, where d_{out} is the hidden-state dimensionality. For the ablation, we vary one hyperparameter at a time while fixing the others to their defaults (i.e., $n = 2000$, $m = 400$, $l = 768$), and report empirical accuracy and false positive rate at significance level $\alpha = 0.01$ in Table 8.

The ablation shows a consistent evidence-scaling pattern: increasing n or l improves identification accuracy while preserving low false positive rates in the reported settings. This matches the theory in Section C.2, where accurate recovery of the population-level bias requires $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$ and γ denotes the bias margin between selected coordinates and the remaining dimensions. Smaller n therefore leaves too little evidence per output dimension, while larger l increases γ through accumulation effects and strengthens separation. Under a fixed, insufficient sequence count, longer random sequences reduce the effective sample burden and improve detection.

The fingerprint-size parameter m has a non-monotonic effect on the margin. If m is too small, too few selected coordinates are available to estimate a stable comparison distribution. If m is too large, irrelevant coordinates add noise and shrink γ , which would require quadratically more samples to recover the same evidence level. Empirically, m in the 200–400 range gives the strongest balance across model sizes, consistent with the heuristic $m \approx 0.1d_{\text{out}}$ for typical $d_{\text{out}} \in [3000, 5000]$.

6 CONCLUSION

We introduced *SeedPrints*, an intrinsic LLM fingerprinting method that grounds lineage evidence in the model’s random initialization rather than in post-hoc training artifacts. The empirical pattern is consistent across the paper: untrained models already exhibit seed-dependent token-preference and representation-bias profiles; same-seed descendants retain statistically significant agreement with those profiles; and different-seed or unrelated baselines do not produce the same evidence. The resulting distribution-correlation test over *identity dimensions* outputs calibrated p-values, enabling fixed-significance lineage decisions instead of relying only on heuristic similarity thresholds. Across LLaMA- and Qwen-style controlled models, OLMo-2-7B pretraining checkpoints, long fine-tuning trajectories, and *LeafBench* deployments involving continued pretraining, instruction tuning, PEFT, quantization, merging, and distillation, SeedPrints preserves practical lineage-detection margins from model birth through later adaptation. These results support SeedPrints as a quantitatively verifiable “Galtonian” fingerprint for provenance and copyright auditing.

Limitations and Future Directions This work isolates fingerprint evidence rooted in initialization-stage bias, but the current method remains a white-box test because it requires access to model

Table 8: Hyperparameter ablation at significance level $\alpha = 0.01$, reporting accuracy and empirical false positive rate for evidence volume (n, l) and fingerprint size (m).

Ablation on n	200	500	2000
ACC	0.7368	0.7778	0.9375
Empirical FPR	0	0	0
Ablation on l	512	768	1024
ACC	0.9048	0.9375	1.0000
Empirical FPR	0.04	0	0
Ablation on m	200	400	800
ACC	0.9375	0.9375	0.875
Empirical FPR	0.04	0	0.1463

540 outputs such as logits or hidden states. It may therefore not apply when only restricted API access is
541 available, especially under deterministic decoding, safety filtering, or limited response visibility. At
542 the same time, the results clarify an important benchmark gap: many existing methods, particularly
543 black-box approaches, rely on signals that emerge later in training or from alignment behavior and
544 therefore provide weaker evidence at the pretrained or early-pretraining stage. Reliable all-stage
545 fingerprints remain urgently needed, especially methods that close the current gap between white-box
546 and black-box verification. See the full discussion in Section [A.2](#).

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A LLM USAGE AND LIMITATION DISCUSSIONS

A.1 LLM USAGE

We use AI assistants, i.e., ChatGPT and Gemini, for writing and formatting support, including grammar checks, clearer quantitative phrasing in figure and table captions, and other surface-level edits. For programming-related tasks, we occasionally use GitHub Copilot and Claude as coding assistants, e.g., for code auto-completion and debugging hints.

A.2 LIMITATIONS AND FUTURE DIRECTIONS

Although this work identifies seed-born bias signals that remain measurable after training, SeedPrints is currently a white-box method. It may not apply when only restricted API access is available, for example when responses are deterministic, safety-filtered, or stripped of logits and hidden states. The empirical results nevertheless show why this limitation is important: most existing fingerprinting approaches rely on training-induced signatures or alignment behaviors and therefore provide weaker quantitative protection for pretrained or early-stage models. By analogy to human fingerprints, which remain stable across age and environment, a true LLM fingerprint should preserve measurable separation across continued training and deployment contexts rather than reflect incidental training artifacts. Achieving this standard under black-box access, where current methods still show a substantial gap relative to white-box evidence, remains a difficult open problem. We hope these results motivate more principled fingerprinting frameworks with calibrated comparisons and robust all-stage margins for large models.

B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

The hypothesis test in Section 4 decides whether two models share lineage by comparing their random-input correlation statistics with a null distribution. The null represents the evidence expected from two *independent models evaluated on random inputs*, so calibration of this distribution directly controls false lineage claims.

We evaluate two approximations to this null: (i) an empirical simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived from the known distribution of Kendall–Tau correlations. This section frames the calibration evidence in three steps: first measuring the true independent-model null, then showing that Gaussian surrogates reproduce its centered shape, and finally validating the analytical approximation used for efficient testing.

B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null is the distribution of correlation statistics for independent models under the same random-input probe. We instantiate independence by comparing untrained models with different random seeds and estimate the null from ~ 2500 such pairs, each using Kendall–Tau correlations over 10,000 random inputs as in Section 3. Figure 3 shows that this empirical true null is Gaussian-like and centered near zero, which is the expected pattern when there is no lineage evidence.

B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Directly estimating the null from independently initialized models is expensive because it requires many model instances and repeated evaluations. We therefore introduce a tractable simulation-based null that uses Gaussian surrogate outputs as no-lineage proxies.

This surrogate is justified by two evidence-preserving properties. First, Gaussian approximations to neural-network outputs are common in auditing and supported empirically and theoretically (Carlini et al., 2022; Lee et al., 2019). Second, the SeedPrints pipeline uses relative ordering and selection operations, such as rank correlations and bottom- m selection, rather than semantic input content. Because independently initialized models have independent parameter matrices, their no-lineage outputs should not contain strong shared structure. Independent Gaussian matrices therefore provide a simple proxy for the null behavior relevant to our statistic.

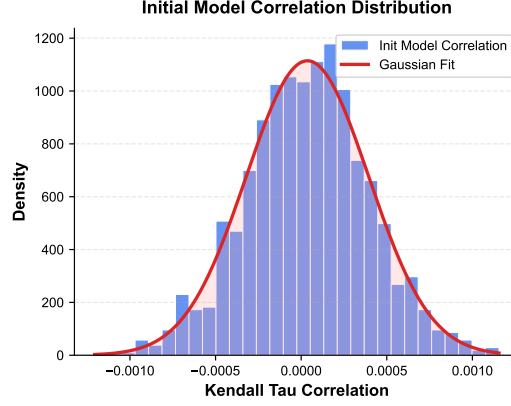


Figure 3: Empirical null over ~ 2500 independently initialized model pairs. Random-input Kendall–Tau correlations concentrate around zero and closely match a centered Gaussian, supporting calibrated no-lineage testing.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape $\mathbb{R}^{N \times d_{\text{out}}}$,
2. Extract the most biased dimensions by selecting the bottom- m ranked average dimensions,
3. Take the intersection set $\mathcal{S}$ and compute correlations on it.

The simulation applies the same selection and aggregation procedure to Gaussian samples, so it preserves dependencies introduced by the pipeline, including those from selecting $\mathcal{S}$ across dimensions. As shown in Figure 4 (middle), the surrogate distribution matches the empirical independent-model null in its centered Gaussian shape, indicating that the simulation provides an effective no-lineage approximation.

B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

Although the Gaussian surrogate captures the full-pipeline null, repeated sampling adds runtime and Monte Carlo variation. We therefore use an analytical approximation as the most efficient calibrated test.

Under the null hypothesis that the two models are independent², each Kendall–Tau correlation τ_j is approximately zero-mean with known variance σ^2 . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization introduces weak dependencies across dimensions. However, Figure 4 shows that the analytical Gaussian closely tracks the Gaussian-surrogate pipeline across settings, including varying numbers of random inputs n and output dimensions D . This match indicates that the dependencies do not materially alter the null variance used for significance testing. A plausible explanation is that the relatively high temperature ($T = 10$) smooths the output distribution and reduces coupling between dimensions.

Together, these results justify the analytical Gaussian null in Section 4: it is efficient, deterministic, and accurate enough to preserve calibrated lineage decisions without constructing costly empirical baselines.

²As discussed in Section B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

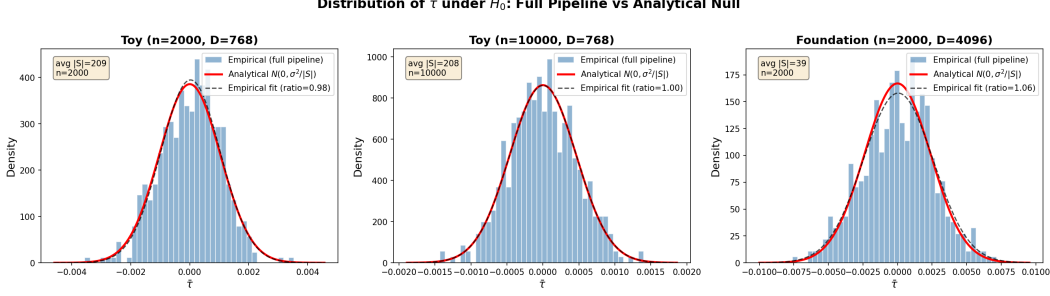


Figure 4: Validation of the analytical Gaussian null. We apply the full pipeline, including identity-dimension selection and correlation aggregation, to Gaussian surrogate outputs and compare the resulting $\bar{\tau}$ distribution with $\mathcal{N}(0, \sigma^2/|S|)$. The close match shows that the analytical approximation captures the no-lineage null despite weak pipeline-induced dependencies.

On variance scaling. The true null from randomly initialized models has slightly smaller variance than the Gaussian null approximation. This variance mismatch does not change our hypothesis-test conclusions. Our primary evaluation for simulation-based null in Section D.2 relies on the Mann–Whitney U-test, which is rank-based and thus invariant to any positive scaling: for any $c > 0$, $x_i < x_j \Leftrightarrow cx_i < cx_j$. Therefore, all rank statistics and rejection thresholds remain unchanged. For tests that do depend on absolute variance, we have additionally evaluated variance-sensitive statistics such as the t -test, whose statistic scales as $t' = t/c$ when variance decreases. Since $c < 1$ in our case, scaling reduces the magnitude of t and makes the test more conservative rather than inflating significance. Empirically, across extensive experiments in Section D.2, we find that the variance scaling does not affect the power of the t -test, indicating that the distributional separation between the alternative hypothesis and the null is substantially larger than the scaled variance.

For the analytical null used in the main paper, variance enters explicitly in the z -score denominator. Any scale discrepancy is therefore absorbed by normalization; because the analytical variance is slightly larger than the empirical variance, the resulting z -scores are smaller, making the test conservative against false positives.

C THEORETICAL PROPERTIES OF OUR METHODS

The theoretical analysis explains why the empirical mean identifies the dimensions that carry the strongest initialization evidence. These are output dimensions that the initialized model is least likely to select under random inputs; because the disfavoring is induced by initialization rather than semantic input distribution, the selected coordinates form the basis of a stable fingerprint signal.

Setup. Consider a model g with fixed parameters and random inputs x_i (uniform random tokens or random embeddings as in the main paper). For each output dimension $j \in [d_{\text{out}}]$, $g_j(x)$ is thus a random variable and its population mean is $\bar{g}_j := \mathbb{E}[g_j(x)]$. The population-level directions we aim to capture correspond to the m output dimensions with the smallest expected values, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the m -th and $(m+1)$ -th smallest means. This gap measures how well-separated the target set $\mathcal{M}$ is from the remaining coordinates. Importantly, for a fixed m and input distribution, $\Gamma(m)$ depends solely on the model’s inherent initialization bias. *However, increasing the length of random input sequences amplifies this bias via concentration, making the separation more pronounced at the population level.*

There are two natural empirical estimators for extracting these population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by the empirical mean over random inputs $X = \{x_i\}_{i=1}^n$ and select the m coordinates with the smallest value. Following the notation in Section 4, where g denotes the model (either the base model f or the suspect model f'):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input $x_i \in \mathbb{R}^{\ell \times d}$, select the smallest-value coordinate and then take the m most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let $p_j := \Pr(j^*(x) = j)$ denote the population probability that index j attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- m set under voting.

Both approaches are intuitively reasonable. However, as we show below, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and can flip under arbitrarily small perturbations, making it inherently noise-sensitive.

C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first show that the sample-mean-based operator, which selects the m smallest coordinates in the empirical mean vector, is stable under small perturbations to model outputs. The perturbations represents realistic variations in outputs from the *same* model, e.g., numerical noise, or evaluation under different precision modes.

Formally, we model perturbed outputs as $g'(x) = g(x) + \eta(x)$, where $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$ represents stochastic deviations. We assume that for each sample x_i and coordinate j , the random variables $\eta_j(x_i)$ are independent across samples (not necessarily across coordinates) and σ -sub-Gaussian with $\mathbb{E}[\eta_j(x_i)] = 0$.³

Lemma C.1 (Stability of sample-mean-based extraction). *Let $\hat{g}$ and $\hat{g}'$ be the empirical means over $X = \{x_i\}_{i=1}^n$, i.e., $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$ and $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$. Let $\widehat{\Gamma}(m)$ denote the empirical gap at rank m . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

Interpretation. The lemma provides a quantitative upper bound on the probability that the sample-mean-based extractor changes its selected set under perturbations. This instability probability decreases exponentially with the number of averaged samples n , due to variance reduction from σ^2 per sample to σ^2/n in the empirical mean.

Proof. We condition on the sample $X = \{x_i\}_{i=1}^n$ throughout the proof, so that $\hat{g}$ and $\widehat{\Gamma}(m)$ are fixed, and all probabilities are taken over the randomness of $\{\eta(x_i)\}_{i=1}^n$ only.

³This sub-Gaussian assumption is mild: bounded random variables are σ -sub-Gaussian (Hoeffding's lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$ is a special case of this assumption.

For any coordinate j , by definition of $\hat{g}'_j$, $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$. Each $\eta_j(x_i)$ is mean-zero and σ -sub-Gaussian, so their average is mean-zero and $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as $1/n$ under averaging. Hence, for any $\varepsilon > 0$, by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than ε :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all d_{out} coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \right\}.$$

Equivalently, if $\max_j |\hat{g}'_j - \hat{g}_j| < \widehat{\Gamma}(m)/2$, then the bottom- m identity set remains unchanged.

Let $j_{(1)}, \dots, j_{(d_{\text{out}})}$ be a permutation of $\{1, \dots, d_{\text{out}}\}$ such that $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$. Recall that the empirical gap at rank m is $\widehat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$. Denote $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$ and $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$. For any $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$ and any $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$, $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \widehat{\Gamma}(m)$. With $|\hat{g}'_i - \hat{g}_i| < \widehat{\Gamma}(m)/2$ and $|\hat{g}'_\ell - \hat{g}_\ell| < \widehat{\Gamma}(m)/2$, we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\widehat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\widehat{\Gamma}(m)}{2}) \geq \widehat{\Gamma}(m) - \widehat{\Gamma}(m) = 0.$$

Thus no coordinate in $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$ can become smaller than any coordinate in $\widehat{\mathcal{M}}^{\text{mean}}(g)$ under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since $\widehat{\Gamma}(m)$ is fixed under conditioning on X , plugging $\varepsilon = \widehat{\Gamma}(m)/2$ into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

In contrast, per-sample voting applies a discontinuous winner-take-all operator on each input. Thus, arbitrarily small perturbations to outputs can change decisions with non-vanishing probability. We still consider the same stochastic perturbation model $g'(x) = g(x) + \eta(x)$.

Lemma C.2 (Instability of per-sample argmin voting). *For a fixed input x , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

denote the argmin and its margin. Then, for any fixed x ,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

and for n i.i.d. inputs $x_1, \dots, x_n$,

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

where $\alpha := \mathbb{E}_x \left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$.

Interpretation. Unlike mean extraction, voting does not benefit from variance reduction: each input incurs a direct hard decision based on noisy outputs. Increasing n only increases the number of opportunities for index-flips.

Proof. We first bound the per-input change probability $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$. Fix x and write $j^* := j^*(x; g)$ for brevity. By definition of the margin, $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$. Let $j^{(2)}$ be an index attaining this minimum, i.e., $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$.

Under the perturbation $g'(x) = g(x) + \eta(x)$, we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since $\eta_{j^{(2)}}(x)$ and $\eta_{j^*}(x)$ are independent $\mathcal{N}(0, \sigma^2)$ variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes, $j^*(x; g') \neq j^*(x; g)$, certainly occurs whenever $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$. Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let $Z \sim \mathcal{N}(0, 1)$. Since $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$, we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore, $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$.

Now consider n i.i.d. inputs $x_1, \dots, x_n$. Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across i , because the perturbations $\eta(x_i)$ are independent across i . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$, and $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$. Taking expectations over the randomness of x_i , we obtain $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$. Therefore, by independence across i ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function $u \mapsto \log(1 - u)$ on $u \in [0, 1)$. Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set $\widehat{\mathcal{M}}^{\text{vote}}$ is determined by the empirical counts of the dimensions $\{j^*(x_i; g)\}_{i=1}^n$. If the empirical gap between the m -th and $(m+1)$ -th most frequent dimensions is of order $O(1/n)$ (i.e., only a constant number of votes), then a change in $j^*(x_i)$ for a single input x_i is sufficient to swap the ordering across this boundary and thus alter the top- m set. Combining this observation with the lower bound $1 - (1 - \alpha)^n$ shows that, in such regimes, the probability that $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$ is bounded away from zero uniformly in n , i.e., per-sample argmin voting does not enjoy variance reduction with more samples. $\square$

C.2 INFLUENCE OF HYPERPARAMETERS

We next quantify how many random inputs the empirical-mean estimator needs to recover the true bias set. We reuse notation from **Setup**: each output coordinate $g_j(x)$ is a random variable under random inputs $x \sim \mathcal{D}$, with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$. $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ represents the sorted means in non-decreasing order, and $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$ is the separation margin. We estimate $\bar{g}$ using n i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimension set is $\mathcal{M}$ with size m , and the empirical estimate is $\widehat{\mathcal{M}}$.

Lemma C.3 (Sufficient condition for recovering the true set). *If $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$ and $2\varepsilon < \gamma$, then the empirical bottom- m set equals the true set: $\widehat{\mathcal{M}} = \mathcal{M}$.*

Proof. For any $j \in \mathcal{M}$ and $j' \notin \mathcal{M}$, we have $\bar{g}_j \leq \bar{g}_{(m)}$ and $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$. The ℓ_∞ bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- m set. $\square$

We next translate this recovery condition into hyperparameter effects. We again adopt the same sub-Gaussian assumption as in Section C.1. This assumption is mild: by Hoeffding's inequality, any bounded random variable is σ -sub-Gaussian with σ proportional to its range. In our setting, each $g_j(x)$ corresponds to a logit or final hidden activation, which are typically bounded or light-tailed in both initialized networks (due to controlled parameter initialization) and trained networks.

Corollary C.4 (Sample complexity under sub-Gaussian coordinates). *Suppose each $g_j(x)$ is σ -sub-Gaussian. Then with probability at least $1 - \delta$,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$.

Proof. For fixed j , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all d_{out} coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to δ and solve for ε to obtain, with probability at least $1 - \delta$,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$ (i.e., if $2\varepsilon < \gamma$), then Lemma C.3 applies and implies $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$. $\square$

Table 9: Cross-seed fingerprint comparisons across model families, testing whether independently initialized models remain statistically separable.

(a) LLaMA-style cross-seed comparisons					(b) Qwen-style cross-seed comparisons				
Seed Pair	Logits Output		Hidden State		Seed Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{42} vs. s_{2000}	0.404	0.456	0.357	0.532	s_{123} vs. s_{1000}	0.741	0.727	0.094	0.074
s_{123} vs. s_{42}	0.214	0.295	0.678	0.565	s_{1000} vs. s_{123}	0.954	0.971	0.125	0.094
s_{1000} vs. s_{123}	0.219	0.246	0.363	0.335	s_{42} vs. s_{2000}	0.273	0.360	0.451	0.529
s_{2000} vs. s_{1000}	0.282	0.291	0.434	0.481	s_{2000} vs. s_{42}	0.215	0.206	0.230	0.295

Table 10: (LLaMA-style models) Same-seed initialization-to-trained comparisons remain significant (p -value < 0.01).

Model Pair	Logits Output		Hidden State		Baselines			
	<i>t</i> -test	<i>u</i> -test	<i>t</i> -test	<i>u</i> -test	Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.33e-3✓	1.02e-3✓	2.20e-8✓	6.28e-8✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{123}^{base}	2.06e-3✓	7.33e-3✓	7.09e-6✓	1.37e-5✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{1000}^{base}	2.44e-3✓	4.14e-3✓	5.58e-4✓	2.81e-3✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{2000}^{base}	5.63e-3✓	6.76e-3✓	4.00e-10✓	1.27e-9✓	-0.337×	0.331×	0.581×	0.188×

Thus accurate recovery requires $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$: the random-query count grows only logarithmically with output dimension but increases quadratically as the separation margin γ shrinks. The margin γ itself has two contributing factors:

- **Intrinsic bias strength.** Longer random input sequences amplify initialization-induced bias via concentration, increasing the population separation between coordinates.
- **Choice of m .** Different choices of m correspond to different parts of the ranked spectrum $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$, thus yielding different effective margins in practice.

The first factor is controlled by input length l , while the second is chosen empirically from the observed gaps in the mean spectrum.

D MORE EXPERIMENT DETAILS

D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all controlled models with the Hugging Face Transformers Trainer (Wolf et al., 2020) and use Accelerate (Gugger et al., 2022) for distributed runs, keeping the implementation consistent across seed comparisons. All open-source models are loaded from official Hugging Face releases and used under their original licenses: Llama models under the Meta Llama Community License and other models under Apache-2.0. All datasets are downloaded via the Hugging Face Datasets library, which is Apache-2.0; dataset content follows each dataset’s stated license.

D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section contains the complementary results for Section 5. We report results using the empirical simulation-based null in Algorithm 1 with both *logits* and *hidden state* outputs, and we test with both the one-sided *t*-test (*t*-test) and Mann–Whitney *U* test (*U* test).

D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-based random baseline is stochastic, we report p -values averaged over 10 independent trials and adopt a significance level

Table 11: (Qwen-style models) Same-seed initialization-to-trained comparisons remain significant (p -value < 0.01).

Model Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test
s_{123}^{init} vs. s_{123}^{base}	2.38e-03	1.68e-03	7.36e-15	3.38e-13
s_{1000}^{init} vs. s_{1000}^{base}	1.35e-04	4.84e-05	4.41e-13	2.01e-11
s_{42}^{init} vs. s_{42}^{base}	1.39e-03	1.46e-03	1.06e-24	2.05e-19
s_{2000}^{init} vs. s_{2000}^{base}	1.80e-03	1.28e-03	4.87e-24	1.92e-20

Table 12: Continual-training comparison on diverse corpora for the seed-1000 openwebtext base model, showing whether each method preserves same-lineage evidence under domain shift. U -test refers to the Mann–Whitney U test.

(a) LLaMA-style model comparisons

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	0 [✓]	0 [✓]	0 [✓]	7.77e-89 [✓]	1.000 [✓]	0.759 [×]	0.999 [✓]	0.996 [✓]
TinyStories (123)	1.000 [✓]	1.00 [✓]	0.943 [✓]	0.902 [✓]	0.950 [×]	0.658 [✓]	0.332 [✓]	0.012 [✓]
the_stack (1000)	0 [✓]	1.73e-287 [✓]	0 [✓]	3.09e-69 [✓]	0.489 [×]	0.557 [×]	0.585 [×]	0.123 [×]
the_stack (123)	0.616 [✓]	0.479 [✓]	0.732 [✓]	0.831 [✓]	0.445 [✓]	0.580 [✓]	0.301 [✓]	0.026 [✓]

(b) Qwen-style model comparisons

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	5.36e-09 [✓]	1.92e-07 [✓]	8.49e-214 [✓]	5.09e-71 [✓]	1.000 [✓]	0.957 [✓]	0.999 [✓]	0.996 [✓]
TinyStories (123)	0.434 [✓]	0.433 [✓]	0.256 [✓]	0.065 [✓]	0.913 [×]	0.199 [✓]	0.328 [✓]	0.039 [✓]
the_stack (1000)	0 [✓]	4.16e-237 [✓]	1.16e-211 [✓]	2.30e-76 [✓]	0.999 [✓]	0.313 [×]	0.995 [✓]	0.976 [✓]
the_stack (123)	0.999 [✓]	0.993 [✓]	0.610 [✓]	0.491 [✓]	0.916 [×]	0.255 [✓]	0.328 [✓]	0.038 [✓]

of $\alpha = 0.01$. Importantly, the absolute magnitude of extremely small p -values is not meaningful: once p falls below numerical and sampling noise (e.g., $< 10^{-20}$), values like 10^{-260} should not be interpreted as stronger evidence than 10^{-20} —both decisively reject the null.

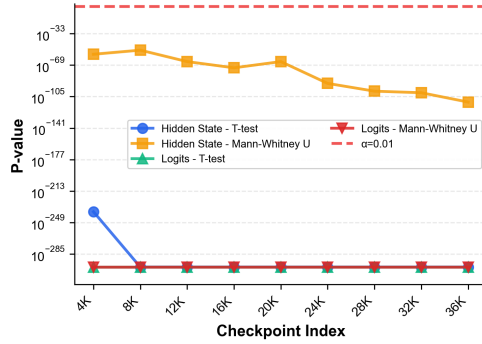
Different initialization seeds produce distinct fingerprints Table 9a and Table 9b report p -values from our correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000) under the t -test and U -test, for Llama-style model and Qwen-style model, respectively. All p -values are consistently > 0.01 , indicating that our method reliably distinguishes models trained from different seeds. This shows that distinct seeds yield distinct fingerprint behaviors, allowing models to be separated “at birth.”

Training preserves the initialization fingerprint. Table 10 (LLaMA-style) and Table 11 (Qwen-style) compares each initialization model s^{init} with its descendant s^{base} trained on the OpenWebText dataset (Gokaslan et al., 2019) (≈ 10 B tokens). Across all seed–model pairs, p -values are consistently < 0.01 , indicating that their bias profiles remain strongly correlated and thus share a common lineage. In short, the trained model inherits the same fingerprint as its initialization; training does not erase the initialization fingerprint. We also evaluate baseline methods in Table 10; without exception, they fail to distinguish across seeds, which in turn suggests their separability stems from training-induced artifacts rather than initialization.

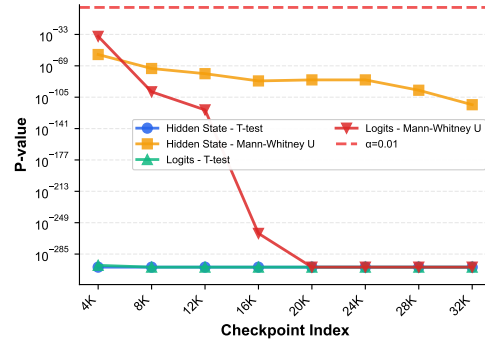
Identical data and order do not make fingerprints converge In Table 13a (LLaMA-style models) and Table 13b (Qwen-style models), all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus (OpenWebText) and in the same data order* (we fix the training seed to lock the data order); the only difference lies in their initialization seeds i . We aim to test

Table 13: The same dataset and training order do not shape fingerprint behaviors to be identical across different initializations.

(a) LLaMA-style models					(b) Qwen-style models				
Model Pair	Logits Output		Hidden State		Model Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{123}^{init} vs. s_{1000}^{base}	0.484	0.500	0.385	0.486	s_{123}^{init} vs. s_{1000}^{base}	0.598	0.638	0.286	0.254
s_{1000}^{init} vs. s_{2000}^{base}	0.946	0.956	0.135	0.096	s_{1000}^{init} vs. s_{123}^{base}	0.804	0.805	0.236	0.240
s_{42}^{init} vs. s_{123}^{base}	0.598	0.589	0.426	0.337	s_{42}^{init} vs. s_{2000}^{base}	0.589	0.608	0.226	0.2043
s_{2000}^{init} vs. s_{42}^{base}	0.756	0.781	0.388	0.287	s_{2000}^{init} vs. s_{42}^{base}	0.523	0.482	0.312	0.323



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: Fingerprint verifies lineage at every checkpoint (p -values < 0.01) across model families.

whether fingerprint behavior would be erased or confounded by identical data and order. Across all cross-seed pairs, p -values remain consistently > 0.01 , in sharp contrast to the near-zero values in Table 10. That is, fingerprints remain seed-specific even under identical data and curriculum.

Continual training on diverse datasets does not confound the fingerprint The purpose of our earlier experiments is solely to demonstrate the strengths of our SeedPrints: it can act as a biometric fingerprint. From a copyright perspective, the inability of prior works to distinguish models with different seeds is not a weakness, since initialization seeds have no clear copyright status. The real fragility is that their attribution can be easily misled by *data distribution, failing to recognize lineage when the training distribution shifts substantially during continued training.*

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two very different datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We compare (i) true descendants trained from the base, versus (ii) *distractors* derived from a different base model (initialized with seed 123 and trained with a different data order on OpenWebText), then continued training on the *same* corpus. The question is whether attribution methods can identify which descendant truly shares lineage with the base.

We find that prior baselines all fail under the code setting (The Stack), misclassifying true descendants as distractors. This indicates that they largely track domain similarity rather than lineage identity: TinyStories is closer in distribution to the pretraining corpus (OpenWebText), while The Stack diverges sharply; such a large distribution shift can easily bypass detection. In contrast, our method correctly attributes lineage across both corpora. Hence, our fingerprint is not a proxy for data distribution: it survives substantial domain shift and persists beyond the initial pretraining stage.

D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. We first conduct a controlled pretraining experiment on OpenWebText, where we know the initialization seed. Each intermediate checkpoint is treated as the base, and we test whether our method can reliably identify its offspring along the same training trajectory. All variants consistently recognize the suspect model as belonging to the same lineage,

Table 14: Cross-size SeedPrints evidence between Qwen-2.5-14B (target) and other Qwen-2.5 models.

	Qwen-2.5-0.5B	Qwen-2.5-1.5B	Qwen-2.5-3B	Qwen-2.5-7B
SeedPrint ($\uparrow$)	0.7184	0.9707	0.3898	0.5429
REEF ($\downarrow$)	0.8250	0.8399	0.8833	0.8461
HUREF ($\downarrow$)	-6.37×10^{-5}	-2.84×10^{-5}	-8.70×10^{-6}	-6.08×10^{-6}

with p -values remaining below the 0.01 threshold (Figure 5a for LLaMA-style models and Figure 5b for Qwen-style models). This confirms the stability of SeedPrints under continuous optimization.

D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To test whether SeedPrints collapses to a family-level signature when model size changes, we conduct a cross-size comparison within the Qwen-2.5 series. Qwen-2.5-14B is the protected target, and Qwen-2.5-{0.5B, 1.5B, 3B, 7B} are suspect models from the same architectural family but not descendants of Qwen-2.5-14B. As in the main paper, SeedPrints outputs a p -value for the hypothesis that the suspect shares initialization-born fingerprints with the target. The baselines, REEF and HUREF, output continuous similarity scores that require threshold selection. Table 14 therefore compares calibrated SeedPrints evidence against similarity-based family-level baselines.

The cross-size results show the desired negative evidence. All SeedPrints p -values remain above a typical significance level such as $\alpha=0.01$, so none of the smaller Qwen-2.5 models is flagged as sharing lineage with Qwen-2.5-14B. This is the favorable outcome for provenance auditing: models can share family design and architecture conventions without being spuriously merged into one fingerprint identity.

The cross-size p -values span different regions of the $[0, 1]$ range rather than collapsing to a single family-level value. We do *not* interpret these magnitudes as a linear “similarity score”; their role is to support a calibrated yes/no decision at a fixed significance level. The spread nevertheless indicates that the statistic remains sensitive to the concrete architecture–initialization combination instead of degenerating into a generic Qwen-2.5 pattern.

The similarity-based baselines show a different evidential profile. REEF stays within a narrow score band (0.8250–0.8833) across sizes, while HUREF produces values close to zero (on the order of 10^{-5}). Because these methods do not provide a built-in decision rule, an operator must choose a threshold and then decide whether cross-size family variants should be separated from true descendants. SeedPrints instead reports a statistical decision at a prescribed significance level, giving a clearer operational answer to whether two same-family models plausibly share training lineage.

E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers 58 models in total. Below we document the decoding configuration used throughout all runs and enumerate the model catalog grouped by family. For each derived model we also indicate the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, *distillation*). “PT” and “IT” refer to *pre-trained* and *instruction-tuned* source models, respectively.

Parameter	Value
max_new_tokens	512
temperature	0.7
top_p	0.9
top_k	50
do_sample	true
max_input_length	512

E.1 MODEL LIST BY FAMILY

E.1.1 QWEN-2.5-7B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-7B	—
IT	Qwen/Qwen2.5-7B-Instruct	—
Derived	Qwen/Qwen2.5-Math-7B	finetune
Derived	Qwen/Qwen2.5-Coder-7B-Instruct	finetune
Derived	WangCa/Qwen2.5-7B-Medicine	finetune
Derived	huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2	finetune
Derived	Locutusque/StockQwen-2.5-7B	merge
Derived	bunnycore/QevaCoT-7B-Stock	merge
Derived	fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4	quantization
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8	quantization
Derived	Lansechen/Qwen2.5-7B-Open-R1-Distill	distillation

E.1.2 QWEN2.5-14B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-14B	—
IT	Qwen/Qwen2.5-14B-Instruct	—
Derived	Qwen/Qwen2.5-Coder-14B	finetune
Derived	oxyapi/oxy-1-small	finetune
Derived	v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	merge
Derived	ToastyPigeon/qwen-story-test-qlora	adapter
Derived	Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4	quantization
Derived	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	distillation

E.1.3 LLAMA-3.1-8B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-3.1-8B	—
IT	meta-llama/Llama-3.1-8B-Instruct	—
Derived	ValiantLabs/Llama3.1-8B-Fireplace2	finetune
Derived	RedHatAI/Llama-3.1-8B-tldr	finetune
Derived	proxectonos/Llama-3.1-Carballo	finetune
Derived	mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated	finetune
Derived	gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES	merge
Derived	XiaoJian9992024/Llama3.1-8B-ExtraMix	merge
Derived	LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching	adapter
Derived	chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7	adapter
Derived	iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8	quantization
Derived	DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ	quantization
Derived	asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill	distillation

E.1.4 MISTRAL-7B-v0.3

Variant	HuggingFace repo	Type
PT	mistralai/Mistral-7B-v0.3	—
IT	mistralai/Mistral-7B-Instruct-v0.3	—
Derived	KurmaAI/AQUA-7B	finetune
Derived	openfoodfacts/spellcheck-mistral-7b	finetune
Derived	grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B	merge
Derived	chaymaemhrioui/mistral-Brain_Model_ACC_Trainer	adapter
Derived	RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit	quantization
Derived	eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b	distillation

E.1.5 GEMMA-2-2B

Variant	HuggingFace repo	Type
PT	google/gemma-2-2b	—
IT	google/gemma-2-2b-it	—
Derived	rinna/gemma-2-baku-2b	finetune
Derived	anakin87/gemma-2-2b-neogenesis-ita	finetune
Derived	vonjack/gemma2-2b-merged	merge
Derived	google-cloud-partnership/gemma-2-2b-it-lora-sql	adapter
Derived	qilowoq/gemma-2-2b-it-4Bit-GPTQ	quantization
Derived	Syed-Hasan-8503/Gemma-2-2b-it-distilled	distillation

E.1.6 LLAMA-2-7B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-2-7b-hf	—
IT	meta-llama/Llama-2-7b-chat-hf	—
Derived	allenai/tulu-2-7b	finetune
Derived	QIAIUNCC/EYE-Llama_qa	finetune
Derived	DevQuasar/coma-7B-v0.1	merge
Derived	Ammar-1/llama2-Better-Tune	adapter
Derived	TheBloke/Llama-2-7B-Chat-GPTQ	quantization
Derived	cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	distillation

E.1.7 TINYLLAMA-1.1B

Variant	HuggingFace repo	Type
PT/IT	TinyLlama/TinyLlama-1.1B-Chat-v1.0	—
Derived	alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2	finetune
Derived	Edentns/DataVortexTL-1.1B-v0.1	finetune
Derived	appvoid/dot-v2.7	merge
Derived	barissglc/tinyllama-tarot-v1	adapter
Derived	TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ	quantization
Derived	anudaw/distilled-code-llama	distillation

Notes. “Type” denotes the post-training transformation used to create a derivative relative to the PT/IT source model: *finetune* includes supervised/preference optimization variants; *adapter* includes LoRA/QLoRA-style modules; *merge* includes model soups and TIES-style merges; *quantization* includes GPTQ/INTx variants; *distillation* includes student models distilled from reasoning-augmented teachers.

1512 E.2 SAMPLE RESULTS

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1514 We present partial results from one LeafBench evaluation run in Table 15 (u -test with simulation-
1515 based null). The reported metric is $1 - p$, used only to align with similarity-score tables. We adopt a
1516 p -value significance level of 0.01; LeafBench’s empirical optimal decision threshold of 0.9920 for
1517 $1 - p$ is consistent with this high-confidence same-lineage region.

1518 Crucially, individual p -values are not linear similarity measures, but their pairwise distribution is
1519 informative: same-lineage comparisons concentrate at much lower p -values, whereas unrelated
1520 comparisons remain near the null behavior.

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Model	Qwen-2.5-7B	Qwen-2.5-7B-Instruct	Qwen2.5-14B	Qwen2.5-14B-Instruct
Qwen-2.5-7B	1.000	1.000	0.079	0.000
Qwen-2.5-7B-Instruct	1.000	1.000	0.813	0.536
Qwen2.5-7B-Math	1.000	0.994	0.047	0.987
Qwen2.5-7B-Coder	0.996	0.999	0.177	0.002
Qwen2.5-7B-Instruct-Medicine	1.000	1.000	0.576	0.369
Qwen2.5-7B-Instruct-Abiliterated	1.000	1.000	0.276	0.886
Qwen2.5-7B-Stock	1.000	1.000	0.939	0.225
QevaCoT-7B	1.000	1.000	0.199	0.119
Qwen2.5-7B-Instruct-Task-10	1.000	1.000	0.672	0.090
Qwen2.5-7B-Instruct-Task-12	1.000	1.000	0.155	0.987
Qwen2.5-7B-Instruct-Int4	1.000	1.000	0.827	0.961
Qwen2.5-7B-Instruct-Int8	1.000	1.000	0.914	0.739
Qwen2.5-7B-Open-R1-Distill	1.000	1.000	0.302	0.955
Qwen2.5-14B	0.324	0.487	1.000	1.000
Qwen2.5-14B-Instruct	0.403	0.702	1.000	1.000
Qwen2.5-Coder-14B	0.492	0.660	1.000	1.000
oxy-1-small	0.492	0.104	1.000	1.000
Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	0.856	0.931	1.000	1.000
Qwen-story-test-qlora	0.112	0.898	1.000	1.000
Qwen2.5-14B-Instruct-GPTQ-Int4	0.583	0.092	1.000	1.000
DeepSeek-R1-Distill-Qwen-14B	0.204	0.000	1.000	1.000
Llama-3.1-8B	0.638	0.714	0.156	0.000
Llama-3.1-8B-Instruct	0.643	0.118	0.128	0.510
Llama-3.1-8B-Fireplace2	0.715	0.532	0.065	0.354
Llama-3.1-8B-TLDR	0.925	0.811	0.331	0.014
Llama-3.1-8B-Carballo	0.001	0.605	0.003	0.946
Llama-3.1-8B-Instruct-Abliterated	0.061	0.005	0.121	0.422
Llama-3.1-8B-Instruct-HalfAbliterated-TIES	0.356	0.509	0.796	0.000
Llama-3.1-8B-ExtraMix	0.640	0.160	0.721	0.875
Llama-3.1-8B-Instruct-cv-job-description-matching	0.684	0.292	0.004	0.001
Llama-3.1-8B-Instruct-PsyCourse-fold7	0.682	0.057	0.063	0.191
Llama-3.1-8B-Instruct-8bit	0.798	0.214	0.154	0.385
Llama-3.1-8B-Instruct-4bit	0.513	0.023	0.005	0.933
Llama-3.1-8B-Instruct-Open-R1-Distill	0.800	0.101	0.772	0.024
Mistral-7B-v0.3	0.224	0.352	0.506	0.044
Mistral-7B-v0.3-Instruct	0.000	0.000	0.984	0.104
AQUA-7B	0.014	0.613	0.703	0.569
Mistral-7B-v0.3-Spellcheck	0.005	0.481	0.872	0.105
Mistral-7B-v0.3-Instruct-demi-merge	0.904	0.774	0.058	0.522
Mistral-7B-v0.3-Brain	0.340	0.983	0.481	0.016
Mistral-7B-v0.3-Instruct-GPTQ-4bit	0.002	0.281	0.089	0.784
Mistral-7B-distilled-from-deepseek-r1-qwen32b	0.009	0.001	0.511	0.015
Gemma-2-2b	0.747	0.869	0.689	0.000
Gemma-2-2b-it	0.944	0.039	0.626	0.060
Gemma-2-baku-2b	0.616	0.231	0.176	0.464
Gemma-2-2b-neogenesis-ita	0.002	0.536	0.079	0.980
Gemma-2-2b-merged	0.951	0.012	0.651	0.075
Gemma-2-2b-it-lora-sql	0.236	0.408	0.000	0.000
Gemma-2-2B-it-4Bit-GPTQ	0.945	0.159	0.000	0.008
Gemma-2-2b-it-distilled	0.482	0.573	0.271	0.276
Llama-2-7b	0.443	0.470	0.148	0.276
Llama-2-7b-chat	0.528	0.868	0.424	0.766
tulu-2-7b	0.125	0.272	0.170	0.888
EYE-Llama_qa	0.657	0.244	0.674	0.968
coma-7B-v0.1	0.953	0.004	0.573	0.731
llama2-Better-Tune	0.136	0.699	0.469	0.860
Llama-2-7B-Chat-GPTQ	0.594	0.603	0.582	0.424
llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	0.349	0.224	0.366	0.976

Table 15: **Sample pairwise evidence.** Rows are target models and columns are base models; cells with $1 - p \geq 0.99$ are highlighted in green to mark high-confidence same-lineage comparisons.